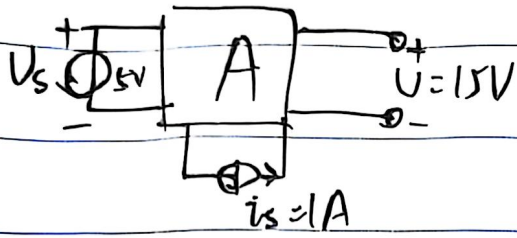


4.30



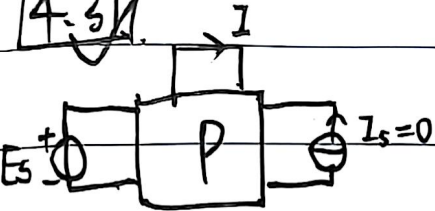
$$U = \alpha U_s + \beta i_s + C$$

$$\text{有: } \begin{cases} 15 = 5\alpha + \beta + C \\ 25 = -5\alpha + \beta + C \\ 5 = -5\alpha - \beta + C \end{cases} \Rightarrow \begin{cases} C = 10 \\ \beta = 10 \\ \alpha = -1 \end{cases}$$

$$\therefore U = -U_s + 10i_s + 10$$

$$U_{15k} = -5 + 10(-1) + 10 = -5V$$

4.31



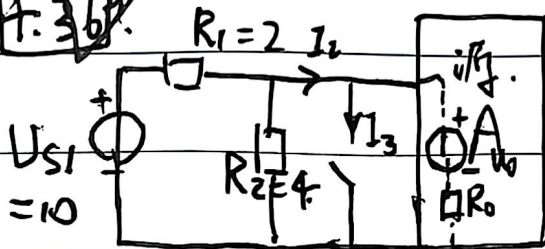
$$I = \alpha E_s + \beta I_s$$

$$4 = \alpha$$

$$\therefore I = 4E_s + \beta I_s$$

$$I' = 12A$$

4.36

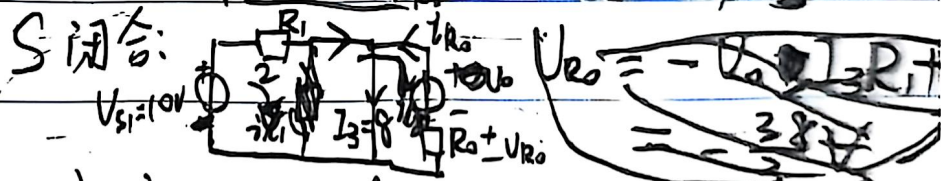
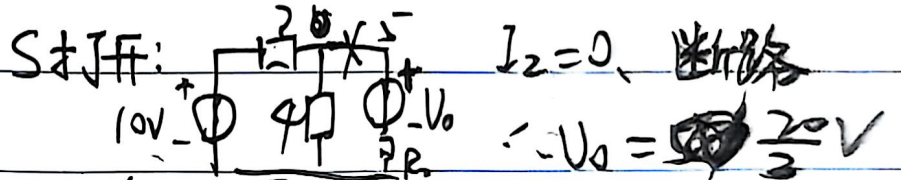


$$i_1 = \frac{U_{s1}}{R_1} = 5A$$

$$\therefore i_{R_0} = I_3 - i_1 = 3A$$

$$R_0 = \frac{U_0}{i_{R_0}} = \frac{20}{3} \Omega$$

$$\begin{cases} U_0 = \frac{20}{3}V \\ R_0 = \frac{20}{3}\Omega \end{cases}$$

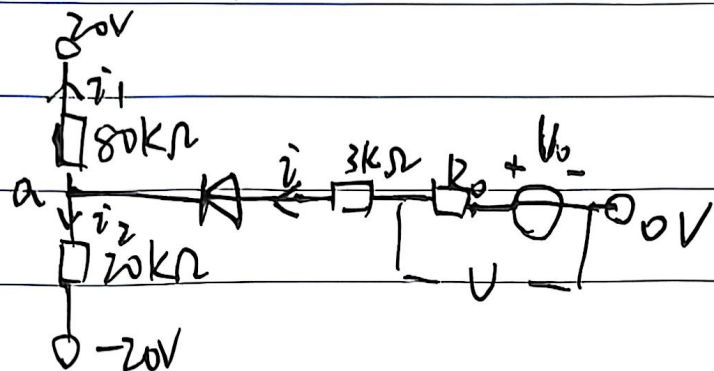


$$i_{R_0} = I_3 = 8A$$

$$i_{U_2} \cdot (R_1 + R_0) + U_0 - U_{s1} = 0$$

4.38

$$u = 4 - 1000i \Rightarrow U_0 = 4V \quad R_0 = 1k\Omega$$



对 a 点:

$$U_a = -20$$

$$U_a - (-20V) = i_2 \cdot 20k\Omega$$

$$U_a - 20V = i_1 \cdot 80k\Omega$$

$$U_a = -i \cdot 4k\Omega + U_0$$

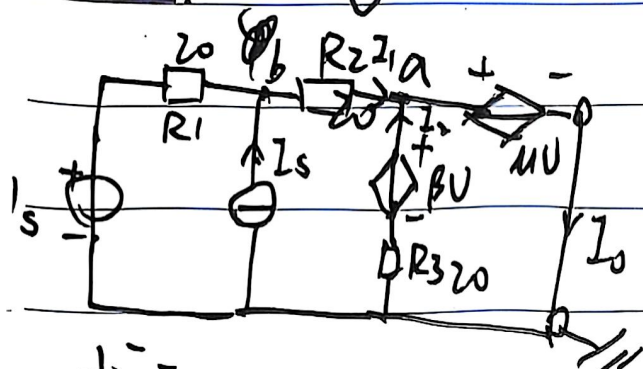
$$i = i_1 + i_2$$

$$\Rightarrow i = 0.8A$$

$$U = U_0 - iR_0 = 3.2V$$

4.39 注意方向

$$Y_a = \beta U - I_2 R_3$$



求 I_0 :

有 $Y_a = 11U$

$$\frac{U_b - U_s}{R_1} + \frac{U_b - U_a}{R_2} = I_s \Rightarrow \begin{cases} U = 12V \\ U_a = 6V \\ U_b = 18V \end{cases}$$

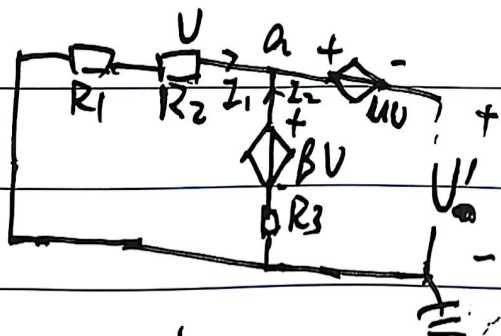
$$U_b - U_a = U$$

$$\therefore I_1 = \frac{U}{R_2} = 0.6A$$

$$I_2 = \frac{U_a - \beta U}{R_3} = 0.15A$$

$$\therefore I_0 = I_1 + I_2 = 0.75A$$

求 R_0 :



$$Y_a = 11U' = \frac{U'}{R_2(R_1 + R_2)}$$

$$\Rightarrow \begin{cases} U = \frac{2}{3}U' \\ U_a = \frac{4}{3}U' \end{cases}$$

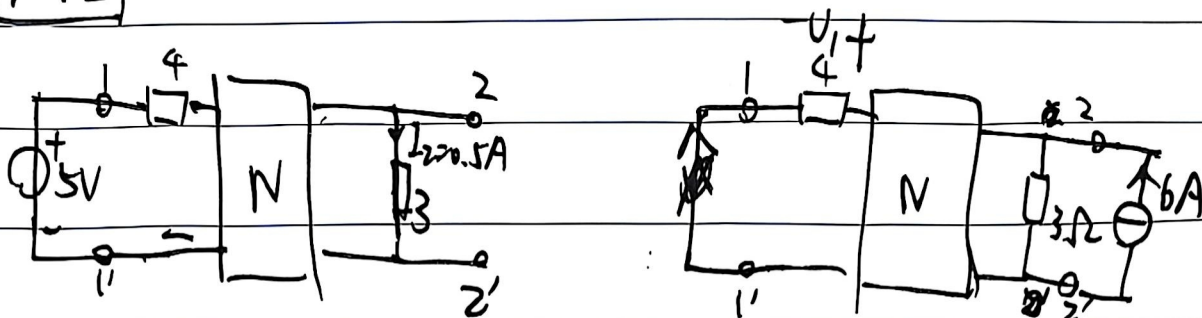
$$\therefore I_1 = \frac{U_a}{R_1 + R_2} = \frac{1}{30}U'$$

$$I_2 = \frac{U_a - \beta U}{R_3} = \frac{1}{24}U'$$

$$\therefore I_{\Sigma} = \frac{30 + 20}{240}U' = \frac{5}{24}U'$$

$$R_0 = \frac{U'}{I_{\Sigma}} = \frac{24}{5} \Omega = 4.8 \Omega$$

4.42

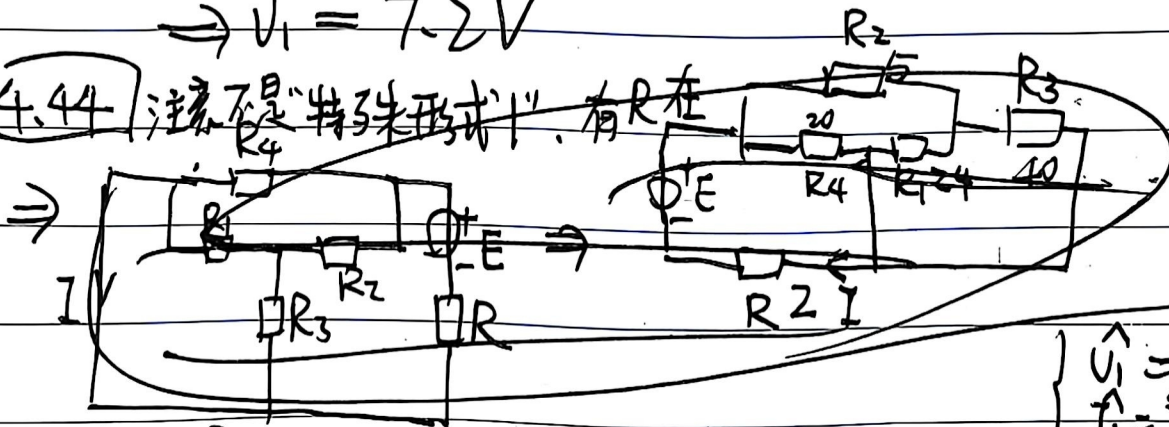


$$5 - \frac{U_1}{4} + 1.5 \times 6 = 0$$

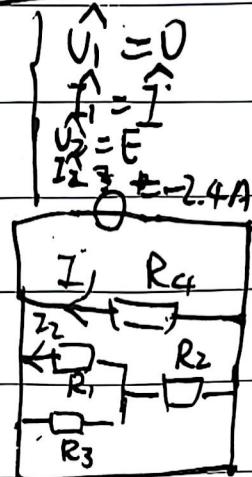
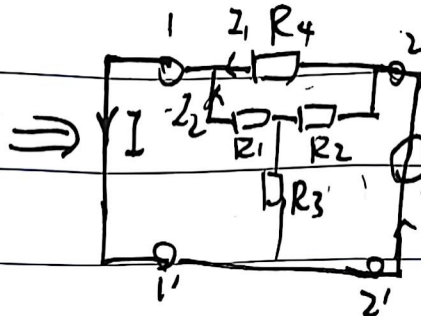
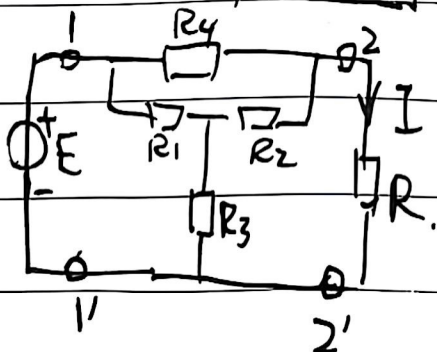
$$\Rightarrow U_1 = 7.2V$$

4.44

注意不是“特殊形式”！有R在



$$\left. \begin{array}{l} U_1 = E \\ I_1 ? \\ U_2 = 2I \\ I_2 = I \end{array} \right\}$$



$$EI = E\hat{I} + 2I\hat{I}E$$

$$\Rightarrow I = 1.625A$$

$$\Rightarrow I = \frac{E}{R_4} = I_1 + I_2$$

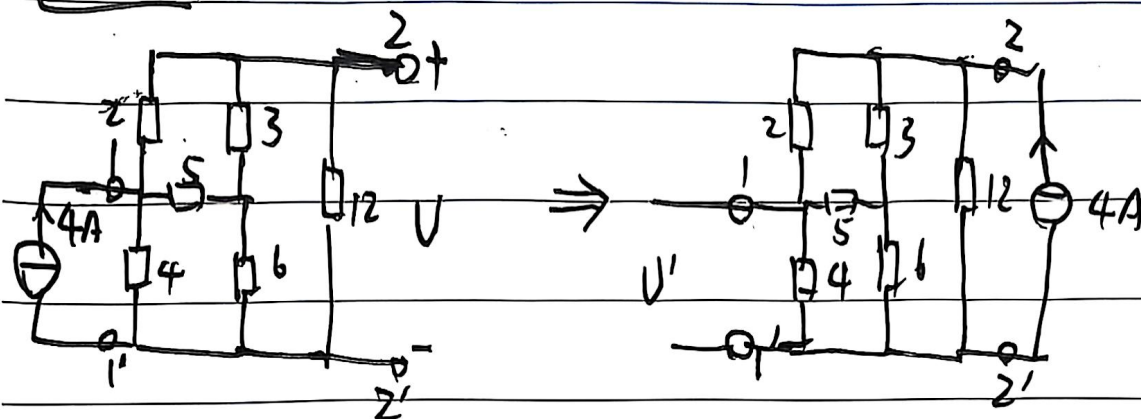
$$I_1 = \frac{E}{R_4} = 1.2A$$

$$I_2 = \frac{E}{R_2 + (R_1 // R_3)} \cdot \frac{R_3}{R_1 + R_3}$$

$$= 0.425A$$

$$\therefore I = 1.95A$$

4.45

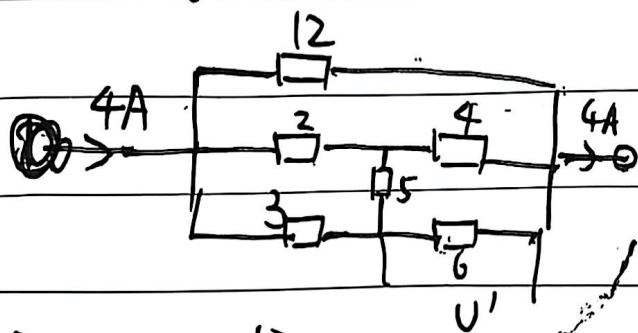


$$\left. \begin{array}{l} I_1 = 4A \\ U_1 = ? \\ I_2 = 0 \\ V_2 = U \end{array} \right\}$$

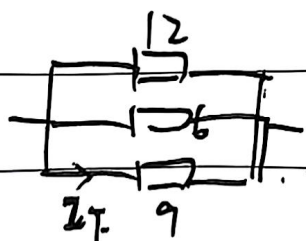
$$\left. \begin{array}{l} \hat{I}_1 = 0 \\ \hat{U}_1 = U' \\ \hat{I}_2 = 4A \\ \hat{V}_2 = ? \end{array} \right\}$$

$$U_1 \hat{I}_1 + V_2 \hat{I}_2 = U \cdot \hat{I}_2 = 4U = \hat{U}_1 I_1 + \hat{V}_2 I_2 = 4\hat{U}_1$$

$$\therefore U = \hat{U}_1 = U'$$



电桥 $\Rightarrow$



$$R_x = \frac{36}{13} \Omega \quad U_x = I R_x = \frac{144}{13} V$$

$$\cancel{I = \frac{4}{13}}$$

$$\therefore U' = \frac{6}{9} U_x = \frac{96}{13} V = U$$